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Interactive Classification Metrics: An interactive graphical Python package for building robust intuition for classification model evaluation

David H. Brown [1¶] **and Davide Chicco** [2,3]

1 Independent Researcher, Austin, Texas, USA **2** Dipartimento di Informatica Sistemistica e Comunicazione, Università di Milano-Bicocca, Milan, Italy **3** Institute of Health Policy Management and Evaluation, University of Toronto, Toronto, Ontario, Canada ¶ Corresponding author

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Summary

Machine Learning continues to grow in popularity in academia, in industry, and is increasingly used in other fields. However, most of the common metrics used to evaluate even simple binary classification models have certain flaws that are neither immediately obvious nor consistently taught to practitioners. Here we present Interactive Classification Metrics (ICM), a Python package to visualize and explore the relationships between different evaluation metrics. The user changes the distribution statistics and explores corresponding changes across a suite of evaluation metrics. The interactive, graphical nature of this tool emphasizes the tradeoffs of each metric without the overhead of data wrangling and model training. The goals of this application are: (1) to aid practitioners in the ever-expanding machine learning field to choose the most appropriate evaluation metrics for their classification problem; (2) to promote careful attention to interpretation that is required even in the simplest scenarios like binary classification. Our software is publicly available for free under the MIT license on PyPI and GitHub.

Statement of need

More people enter the machine learning field every year through a variety of paths. In the United States alone, "Data Scientist" job growth is expected to outpace the average over the next decade, with similar growth in other countries (Bureau of Labor Statistics, 2024; Kaggle, 2022). User-friendly, low barrier to entry libraries like scikit-learn are used widely across the industry, and fields beyond statistics increasingly use machine learning tools, but practitioners are not always familiar with basic best practices like avoiding data leakage, not to mention careful interpretation of evaluation metrics (Chekroud et al., 2024; Kaggle, 2022; Kapoor & Narayanan, 2023; Liu et al., 2019).

Classification models are a fundamental tool in machine learning. The quality of a classification model is evaluated by comparing model predictions with ground truth targets, forming sections of the confusion matrix, and resulting in the True Positive Rate, True Negative Rate, Positive Predictive Value, and Negative Predictive Value (NPV). From these four "basic rates" (Chicco & Jurman, 2023), further evaluation metrics have been derived, each summarizing different aspects of a model's predictive performance. Common metrics have specific caveats known to experts, but not immediately apparent to novices, and can even be overlooked by experienced modelers. For example, interpretation of Accuracy, Recall, and most other metrics depend on class (im)balance and comparison to each other. While these metrics do not claim to guarantee that all four "basic rates" are high, in practice they are often reported on their own or as a single number that adequately describes a model's predictive quality.

Literature details the shortcomings with Receiver Operating Characteristic (ROC) curves (Chicco & Jurman, 2023) and singular metrics (Powers, 2020); has evaluated less common metrics like bookmaker informedness and Matthews Correlation Coefficient (MCC) (Chicco et al., 2021; Chicco & Jurman, 2023); and even proposed novel graphical tools like the MCC-F1 curve (Cao et al., 2020). However, insights from scholarly work take time to enter widespread educational material, and are not immediately obvious from static examples or mathematical formulae alone.

It is essential that the machine learning community have every pedagogical resource at its disposal to solidify a deep understanding of and intuition for the field's most basic, but nuanced, model evaluation tools. ICM aims to improve the choices novice scientists make, and solidify those of experienced practitioners, when building classification models.

Overview of Functionality

After installing ICM in a Python environment, it is callable from the terminal with a single command, which opens a web browser for interaction. It runs on the user's local machine using bokeh server and standard data/machine learning Python libraries for underlying computation.

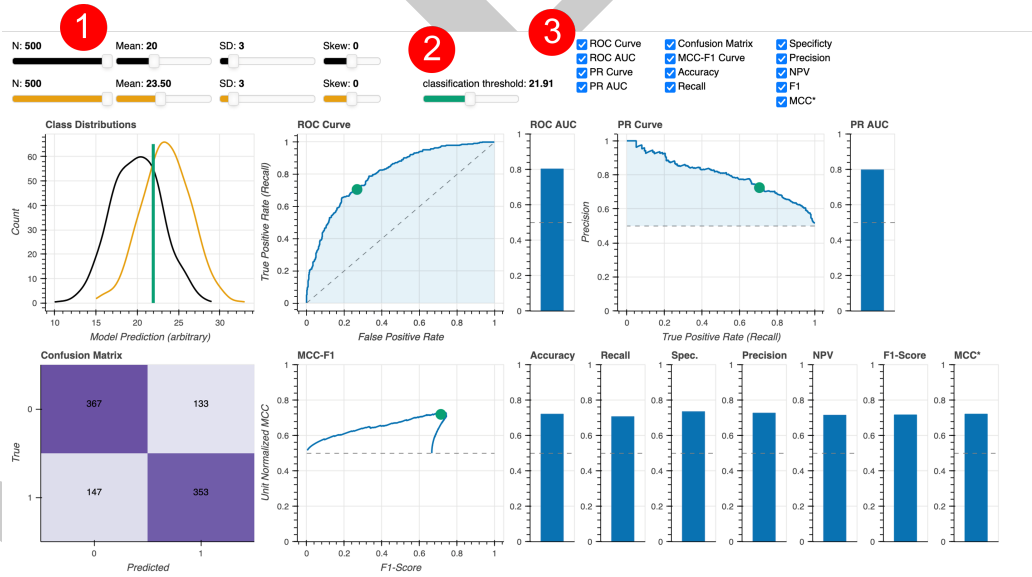

Figure 1: Screenshot of the application with numbered steps overlaid (red circles). Users control 9 interactive sliders at the top, and all graphs respond accordingly. The sliders control the sample size (N), mean, standard deviation (SD), and skew of the two distributions (Step 1) that represent the negative (black) and positive class predictions (orange). The properties of these distributions, along with the classification threshold (green; Step 2), control the magnitude and shape of all other plots. Users can also choose to show or hide specific plots with checkboxes (Step 3). Full display shown.

A brief animation of the application in use can be seen on GitHub. The full list of plots, evaluation metrics, and their acronyms are: Class distributions, ROC curve, Area Under the ROC curve (ROC AUC), Precision-Recall (PR) curve, Area Under the PR curve (PR AUC), Confusion Matrix, Matthews Correlation Coefficient-F1 (MCC-F1) curve, Accuracy, Recall, Specificity, Precision, Negative Predictive Value (NPV), F1-Score, and MCC (unit normalized). Different fields have different naming conventions for the same metric; for example, Recall is also called the True Positive Rate, Sensitivity, or Hit Rate. A comprehensive list of alternative naming conventions is on the confusion matrix Wikipedia page.

Example Scenario

Figure 2 depicts ICM demonstrating a classic model evaluation mistake: using Accuracy alone to evaluate an overfit model trained on an imbalanced dataset. A comprehensive look at all available metrics reveals the flaw: with additional information beyond Accuracy and ROC AUC, it is clear that when accounting for all four "basic rates", MCC reveals chance performance (MCC = 0.5). Also note that PR AUC=0.9 could be naively interpreted as a good model, but its minimum baseline, driven by the proportion of classes, is 0.83, indicating that overall the area covered by the PR curve is small.

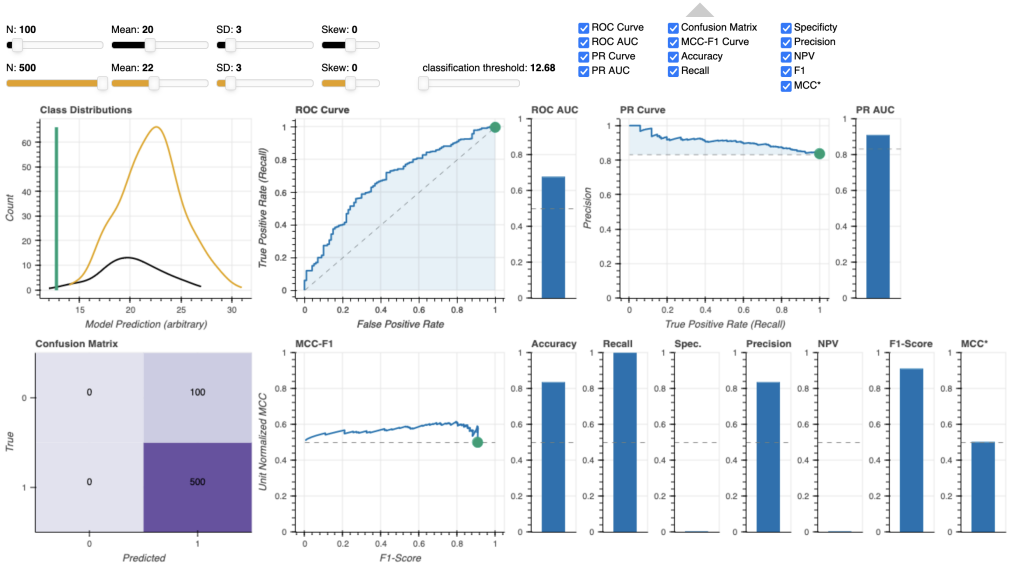

Figure 2: The classic flaw of Accuracy on an imbalanced dataset. The negative class (black) has N=100 examples, the positive class (orange) has N=500. The classification threshold (green) is set extremely low to represent a model that predicts everything as the positive class, yet achieves over 80% Accuracy due to the proportions of the two classes in the dataset.

Conclusion

Visualizing results is a pivotal task in many fields, and scientific visualization is a cornerstone of modern data science (Midway, 2020). Software tools can make this task easier for researchers and students, but to the best of our knowledge, no Python package for visualizing binary classification results in an interactive, exploratory way exists. We fill this gap by providing our ICM software. Our application does not require the overhead of cleaning data and training models. Instead, it allows users to focus on understanding the evaluation metrics themselves, and how these metrics change in relation to each other as the user alters the statistics of the two distributions.

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Conflict of Interest

The authors declare no conflict of interest in this work.

At the time of writing DHB is employed by an artificial intelligence company. The software described in this paper was developed independently by DHB as a personal project. His employer is not affiliated with the development of the software, has no ownership rights, and did not provide any resources or support for this work. The author's contributions to the project were made entirely outside of their employment with any entity.

Author Contributions

DHB: conceptualization, software implementation, application design, writing

DC: application design, domain expertise, scholarly integration, writing

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